JOÃO VITOR CAVALCANTI DA SILVA

Full Stack Developer

Caruaru, Pernambuco — Brazil joaovitorrkc@gmail.com +55 (81) 98212-0328 joaovitorkc.com.br GitHub @joaovitorkc
LinkedIn /in/joaovitorkc

SUMMARY

Full stack developer. I build and maintain the public healthcare ecosystem now running across more than 400 Brazilian municipalities in 26 states — from database modeling and the data pipeline through to the interface, the production servers and project delivery. I also ship consumer products where the hard part is integrity: finite seats, payments, and draws that have to be auditable. TypeScript end to end.

IMPACT

400+	26	200+	50+	300+
MUNICIPALITIES SERVED	STATES	SCREENS IN PRODUCTION	API MODULES	DATABASE MIGRATIONS

EXPERIENCE

FULL STACK DEVELOPER · PUBLIC HEALTH SYSTEMS

Mar 2023 — present

CLIENT Wi Consultoria · Caruaru, PE

- I build and maintain the e-Gestão ecosystem — seven systems under a single portal (R-SUS, SUS Direto, Visa, BPA Online, Call Panel and an AI assistant), in daily use inside municipal health departments across more than 400 municipalities in 26 states. I run the ongoing projects: prioritisation, scope and delivery organisation.
- Designed the pipeline that extracts local e-SUS APS databases, transforms and materialises them into PostgreSQL mirror tables, with an observable Bull/Redis queue. It took read load off the city's health record and made every report reproducible.
- Implemented multi-tenant isolation by municipal IBGE code, resolved and verified server-side — a non-negotiable requirement for health data from hundreds of cities on one installation.
- Removed an entire class of bug by migrating PDF generation from a legacy generator in another language to a print route captured with Puppeteer: screen and document now share a single source of truth.
- Integrated real-time patient calling into the federal health record without changing a line of it — a browser extension, a Fastify WebSocket pub/sub server, and the reception TV with speech synthesis.
- PostgreSQL modeling and query tuning, a schema versioned across 300+ migrations, and hardened deployment on Linux, Nginx and PM2 with the API separated from the queue worker.

STACK TypeScript · Next.js · React · Express · Fastify · PostgreSQL · Redis · Bull · Socket.IO · Puppeteer · Linux · Nginx

FULL STACK DEVELOPER · WEB PRODUCTS

CLIENT Catsuc Labs · Remote

- Quota-based roulette system: each chip purchased is a numbered seat, guaranteed by a unique constraint and a transactional reservation — the database refuses the double sale instead of the application hoping the race never happens.
- Idempotent payment confirmation keyed per event, treating webhook redelivery as the expected case, and a fire-once draw with seed and result persisted so it can be reconstructed afterwards.
- A football pool with deadlines enforced server-side (never on the client clock) and scoring as a pure function of prediction, result and tournament rule — which made recomputation safe when an official result was corrected.
- Dodreams: modeling a subjective domain as a small core with optional fields and free-form steps, with progress derived from completed steps rather than editable by the user.

STACK TypeScript · Next.js · React · Node.js · PostgreSQL · Prisma · Redis

SYSTEMS IN PRODUCTION

Proprietary client code. Full write-ups of the main ones at joaovitorkc.com.br.

E-GESTÃO	The hub of a public healthcare ecosystem	WI CONSULTORIA	TYPESCRIPT · NEXT.JS
PAINEL DE CHAMADAS	Real-time patient calling	WI CONSULTORIA	TYPESCRIPT · REACT
ROLETA ONLINE	Quota-based draw with limited seats	CATSUC LABS	TYPESCRIPT · NEXT.JS
BOLÃO DE FUTEBOL	Predictions, scoring and ranking	CATSUC LABS	TYPESCRIPT · NEXT.JS
DODREAMS	From dream to plan	CATSUC LABS	TYPESCRIPT · NEXT.JS
R-SUS	Primary-care reporting	WI CONSULTORIA	TYPESCRIPT · NEXT.JS
SUS DIRETO	Citizen portal	WI CONSULTORIA	TYPESCRIPT · NEXT.JS
VISA	Municipal health surveillance	WI CONSULTORIA	TYPESCRIPT · NEXT.JS
BPA ONLINE	Outpatient production to the official file	WI CONSULTORIA	TYPESCRIPT · NEXT.JS
RAID	Inherited clinical management, kept alive	WI CONSULTORIA	JAVASCRIPT · REACT

TECHNICAL SKILLS

LANGUAGE	TypeScript · JavaScript · SQL · Python · C · PHP
INTERFACE	React · Next.js · Tailwind CSS · TanStack Query · shadcn/ui · Radix · React Hook Form · Zod · Recharts · Vue.js · GSAP
SERVER	Node.js · Express · Fastify · Knex · Prisma · Bull · queues · WebSocket · Socket.IO · REST · Puppeteer
DATA	PostgreSQL · Redis · MySQL · Migrations · ETL · sync
AI & AUTOMATION	Agent harness · Claude Code · Cursor · Composer · Context engineering · RAG · Self-hosted LLM · Ollama · Process automation
INFRA & OPS	Linux · VPS · Nginx · PM2 · Docker · AWS · S3 · Google Cloud · Git · CI/CD · VPN · hardening

EDUCATION

Computer Science — Jan 2025 — present

UniFavip Wyden

Degree in progress. Data structures in C, algorithms and complexity — the fundamentals practice leans on daily without ever naming them.

CERTIFICATIONS AND COURSES

- **2025** Cloud Architecting — AWS Academy
- **2025** Cyber Threat Management — Cisco
- **2023** JavaScript Intensive — Hashtag Treinamentos
- **2022** Complete Web Development — 6 projects — Udemy
- **2025** Cloud Foundations — AWS Academy
- **2025** Full Stack Week — Full Stack Week
- **2022** React from Zero to Mastery — hooks, router, API — Udemy

COMPANY AND CLIENTS

Quiral Labs — own company

Wi Consultoria — Public health software

Catsuc Labs — Custom software studio

LANGUAGES

Portuguese — Native

English — Intermediate